

Malaviya National Institute of
Technology, Jaipur

Predicting Mileage: Harley Davidson X440

A Machine Learning Approach to Fuel Efficiency

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Dept. of Mechanical Engineering

HARLEY- DAVIDSON® X440



METHODOLOGY & DATA COLLECTION STRATEGY

INITIAL VARIABLES TRACKED

- Load (kg)
- Traffic Conditions
- Hard Throttles
- Road Conditions
- Hard Braking
- Gear Changes
- Average Speed



1. RENT & TEST RIDE
(Full Day)



2. MEASURE & FILL FUEL
Precisely



3. RIDE UNTIL EMPTY
(Run-Dry Method)



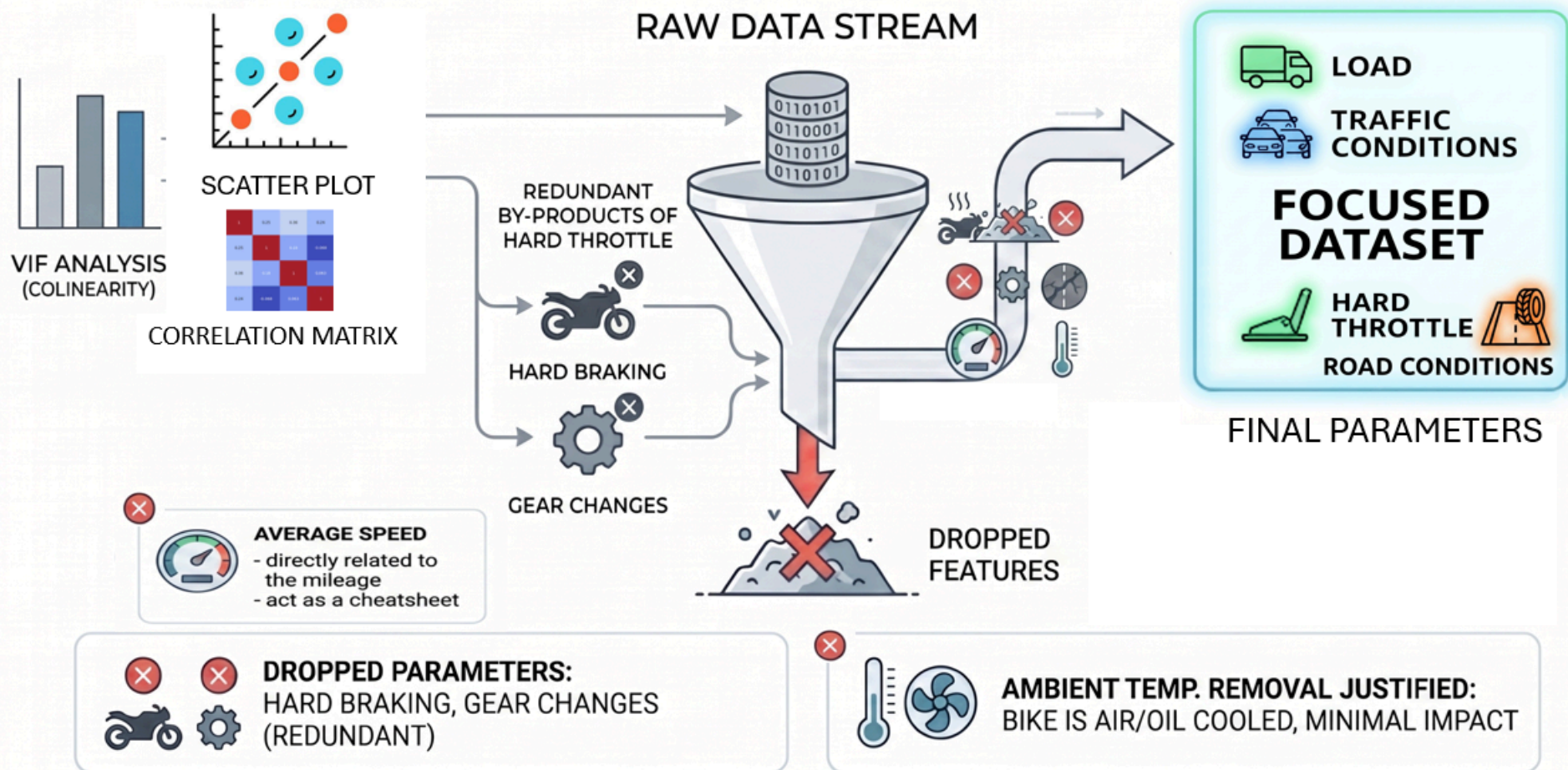
4. GATHER AUTHENTIC
GROUND-TRUTH DATA

CREDITS: PROF. DILIP SHARMA



STRAVA

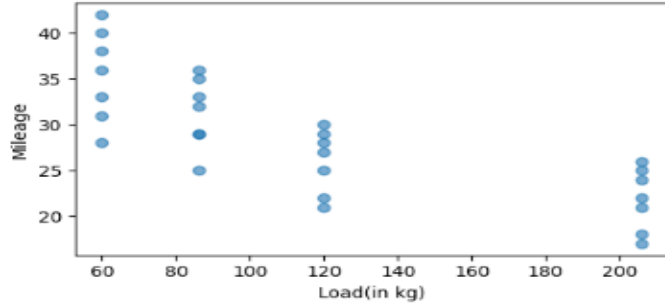
FEATURE SELECTION STRATEGY



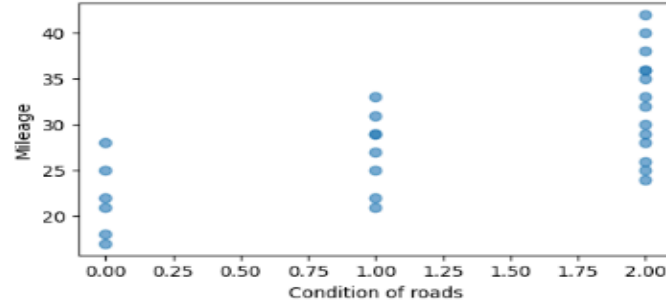
SCATTER PLOTS

Feature vs Mileage Scatter Plots

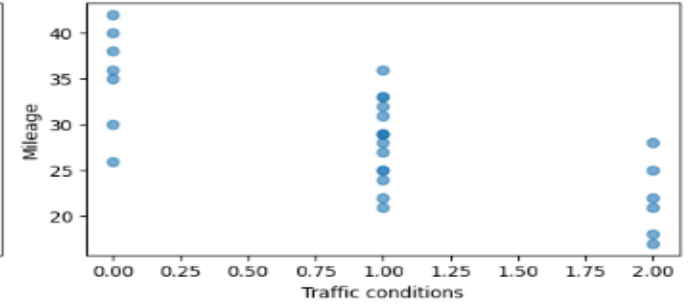
Load(in kg) vs Mileage



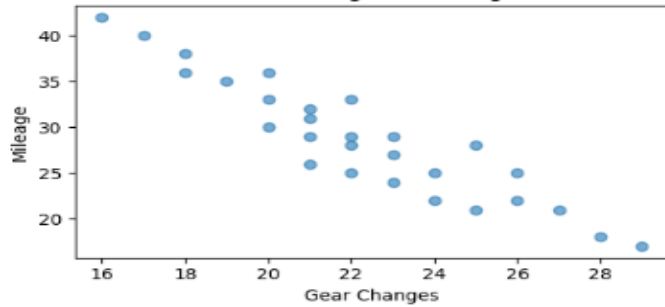
Condition of roads vs Mileage



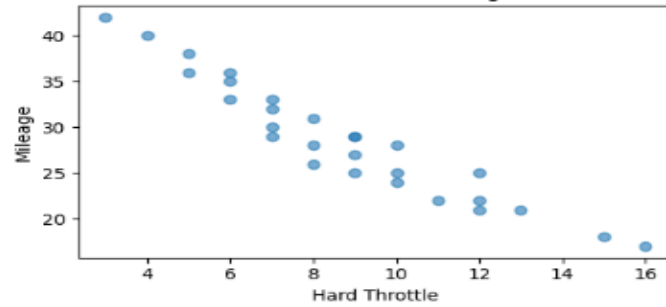
Traffic conditions vs Mileage



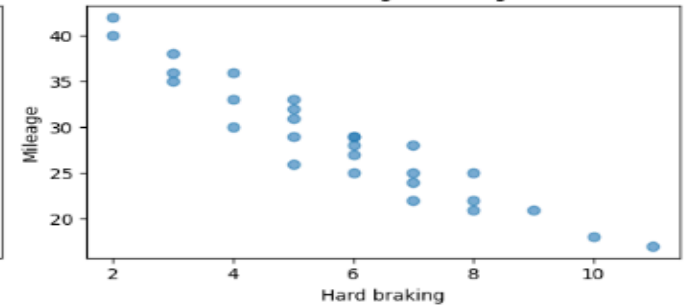
Gear Changes vs Mileage



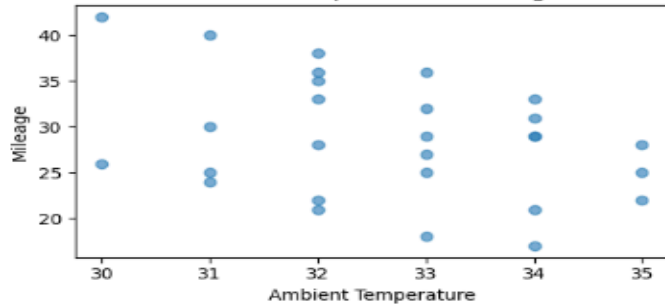
Hard Throttle vs Mileage



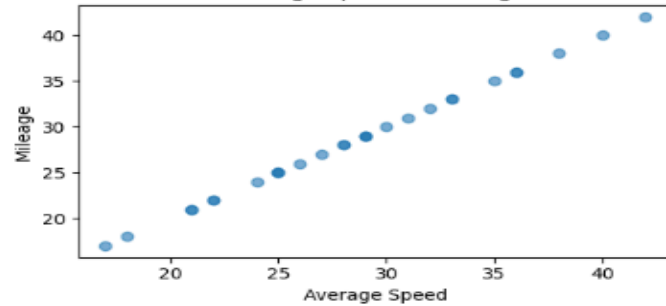
Hard braking vs Mileage



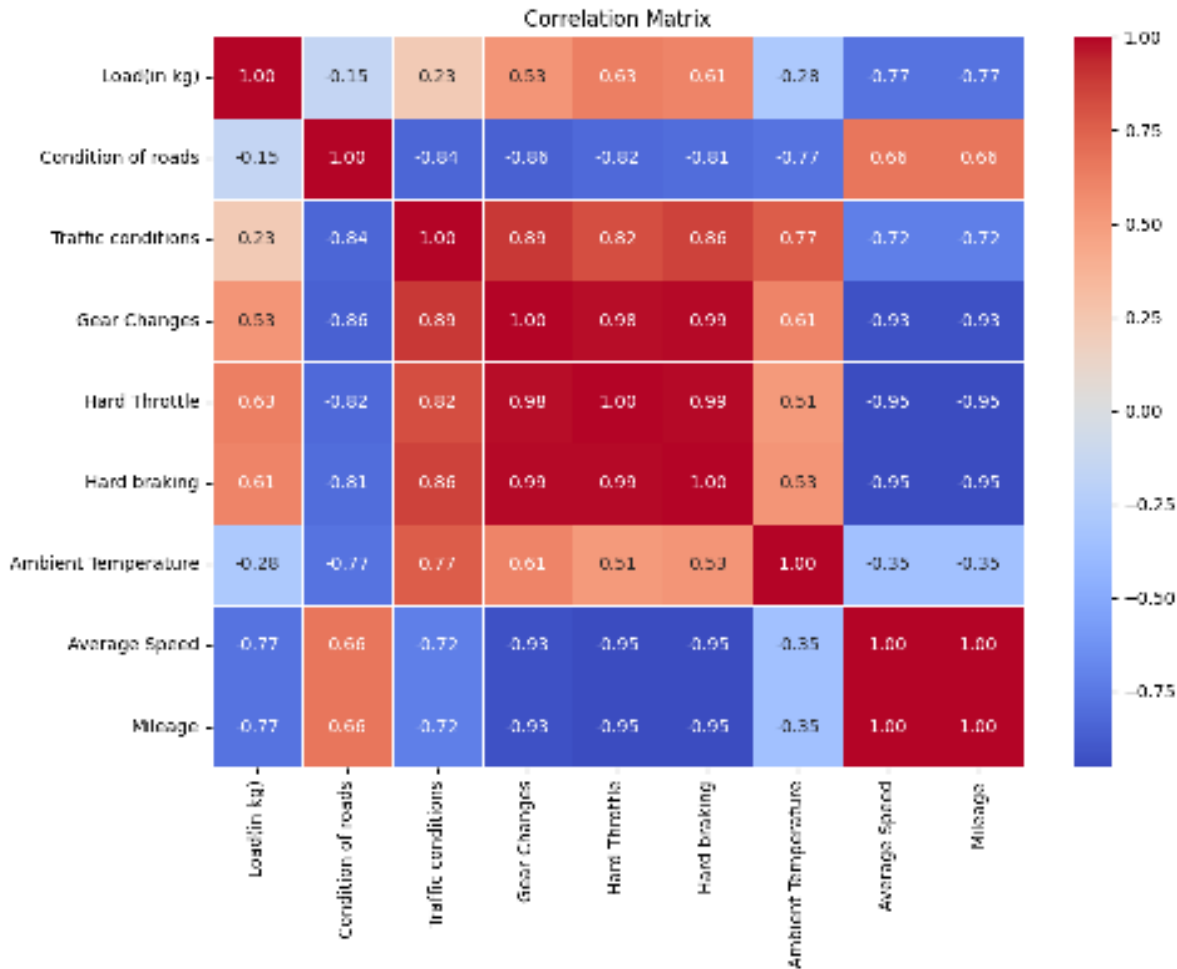
Ambient Temperature vs Mileage



Average Speed vs Mileage



CORRELATION MATRIX



VIF SCORES

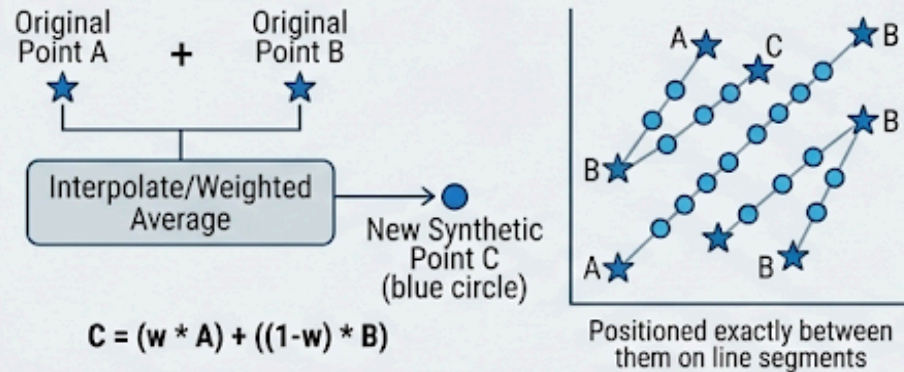
VIF Scores:

Feature	VIF
Gear Changes	109.079864
Hard braking	105.484160
Hard Throttle	101.440954
Average Speed	41.499085
Load(in kg)	22.440826
Condition of roads	11.498448
Traffic conditions	11.173460
Ambient Temperature	10.548910

DATA AUGMENTATION: EXPANDING THE FEATURE SPACE

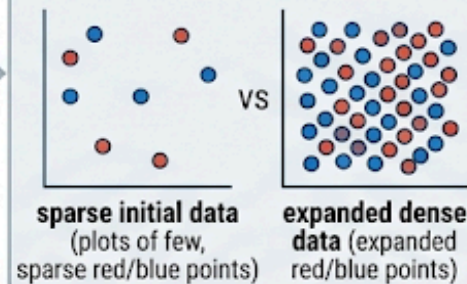
Synthetic data generation to enhance model robustness.

THE METHOD: INTERPOLATION-BASED AUGMENTATION

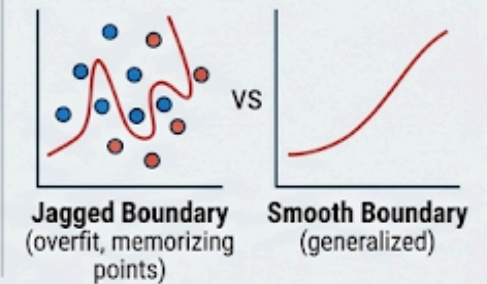


IMPORTANCE

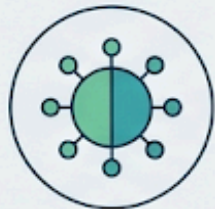
Overcoming Data Scarcity.



Smoothing the Decision Boundary: prevents model memorization.



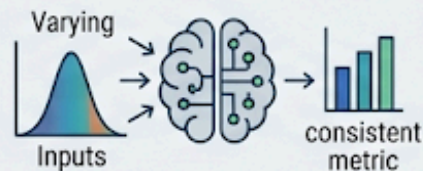
ADVANTAGES



Improved Generalization:

Model performs better on unseen examples and intermediate cases.

Stability:

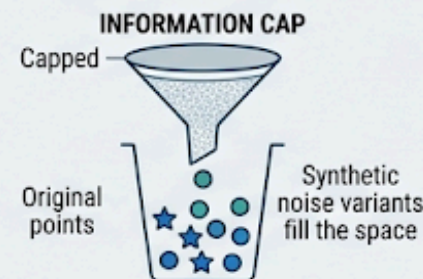


Stability:

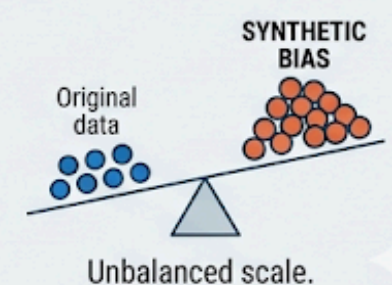
Consistent performance across varied input distribution.

LIMITATIONS

Information Cap.

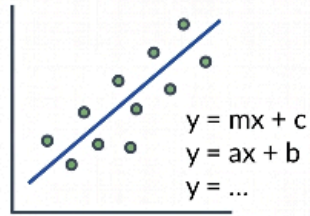


Risk of Synthetic Bias.



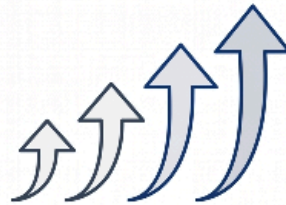
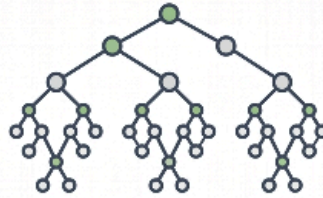
OUR PROGRESSIVE MODELING APPROACH

**BASELINE:
LINEAR MODELS**



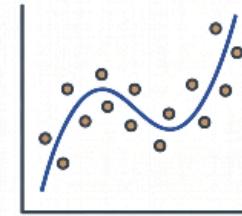
REGRESSION:
Multi-Linear,
LASSO, RIDGE

**COMPLEX:
NON-LINEAR ENSEMBLES**



TREE MODELS:
Random Forest,
GRADIENT BOOSTING

**COMPARISON:
PHYSICAL MODEL**



CURVE FITTING

UNDERSTANDING BASE MODELS: REGRESSION MODELS

THE CORE IDEA



ESTABLISH A BASELINE:
TEST IF MILEAGE SCALES
LINEARLY WITH OUR FEATURES.



LOAD



TRAFFIC
CONDITIONS

**FOCUSED
DATASET**



HARD
THROTTLE

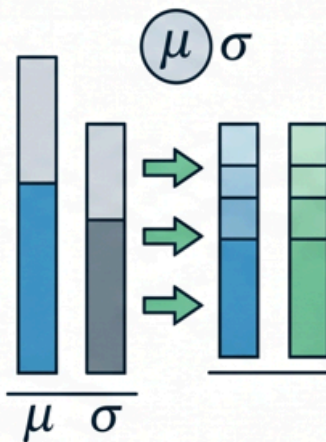
METHODS USED

FEATURE PRUNING



SELECT ONLY RELEVANT
PREDICTORS

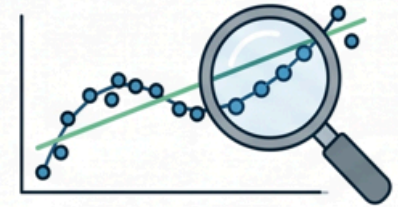
STANDARDIZATION



SCALE FEATURES TO
UNIFORM RANGE (E.G.,
Z-SCORE NORMALIZATION)

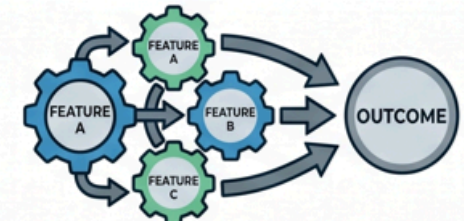
THE LIMITATIONS

THE LINEARITY TRAP



ASSUMED LINEAR RELATIONS.
MODEL MAY MISINTERPRET CURVED DATA.

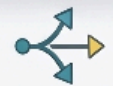
MULTICOLLINEARITY



STRONG DEPENDENCIES BETWEEN
FEATURES CONFUSE COEFFICIENT
INTERPRETATION.
LEADS TO UNSTABLE PREDICTIONS.



Tree-Based Ensembles: Random Forest & Gradient Boosting



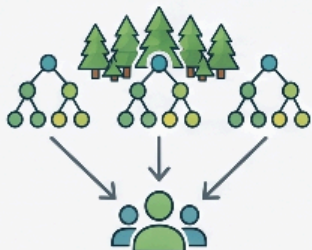
Sub- hierarchical decision-making.

THE CORE IDEA

These models split the data into “decision branches”



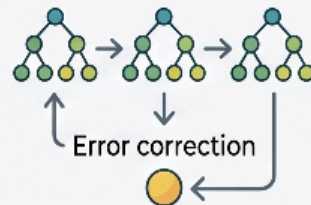
RANDOM FOREST



VOTING MODELS

Bagging multiple trees

GRADIENT BOOSTING



CORRECTING MODELS

Sequential boosting

METHODS USED

NON-PARAMETRIC ADVANTAGE



Tree models are naturally robust to outliers

CATEGORICAL HANDLING



Categorical Handling

THE CHALLENGES & PROBLEMS

1 DATA HUNGRY MODELS



Data hungry models

Needs lots of data to generalize

2 OVERFITTING RISK



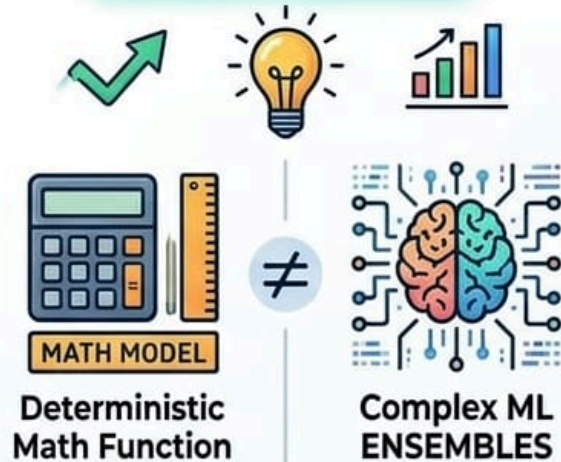
Overfitting Risk

Can “memorize” the 28 training points.
Contrast: Gradient Boosting is better at finding the “signal” without overfitting the “noise” in small datasets.

Traditional Curve Fitting Analysis

Validating Machine Learning vs. Pure Mathematical Modeling.

The Core Idea

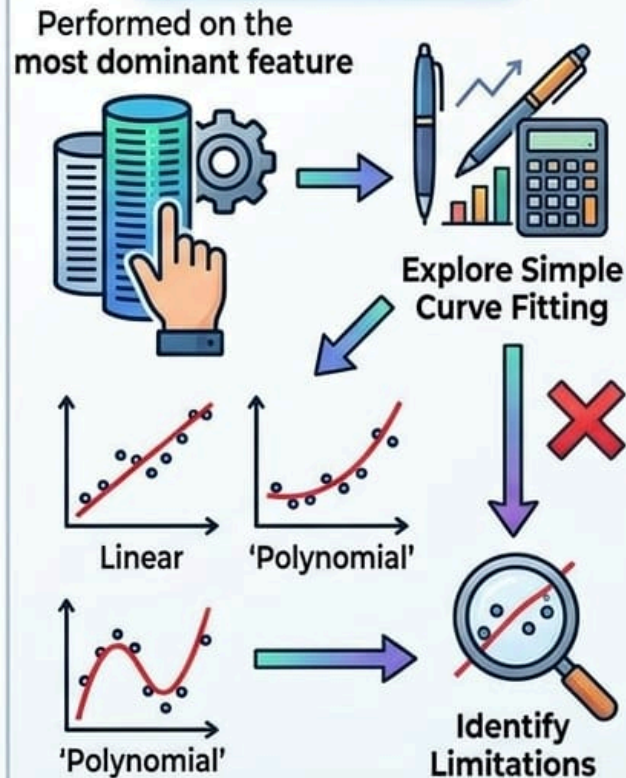


The Core Idea:

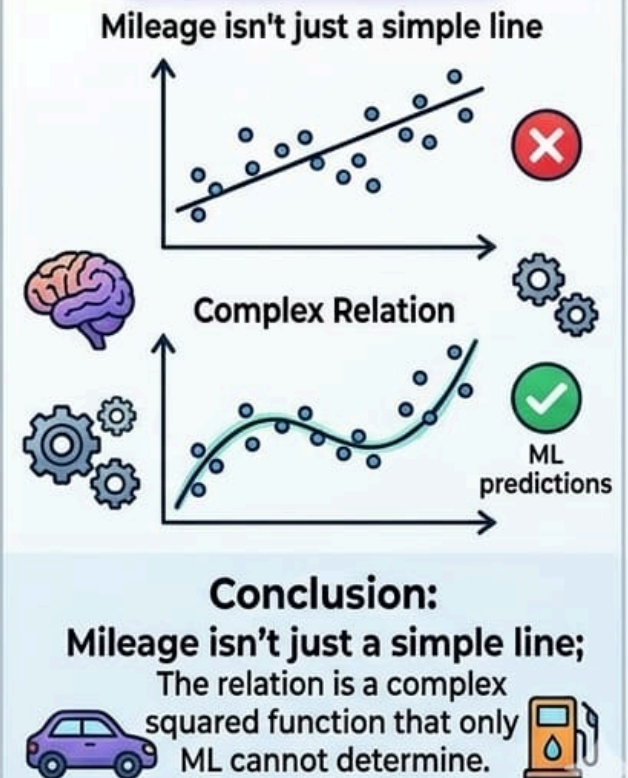
To determine if a simple, deterministic mathematical function could explain the mileage variance as effectively as complex ML ensembles.

Fit to model the primary trend.

Data Strategy



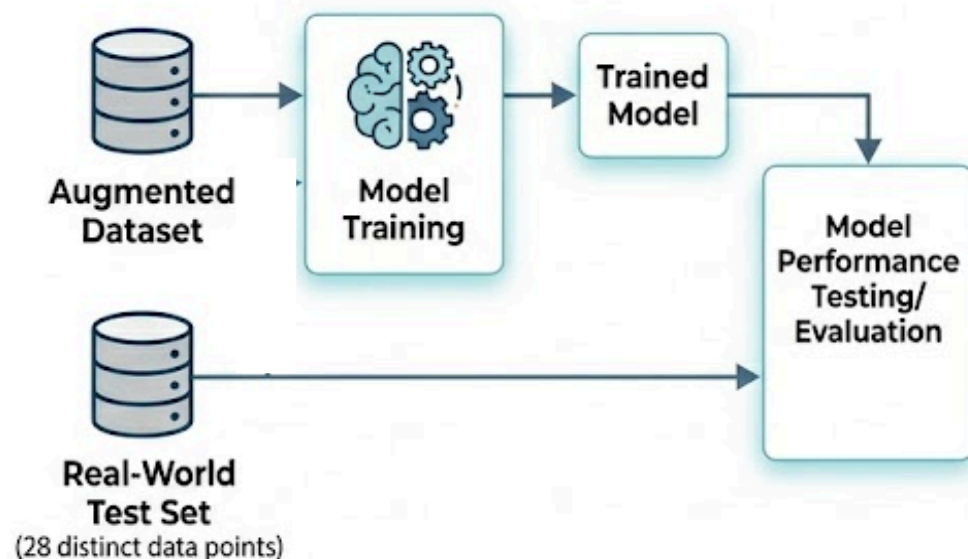
Conclusion



Model Training & Validation

From pattern learning to rigorous performance verification.

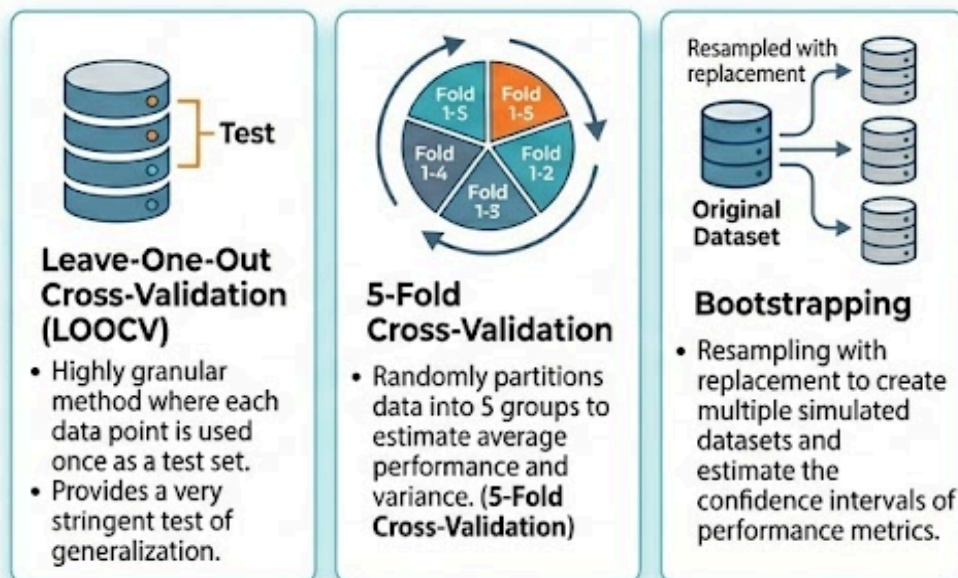
1. Model Training Strategy



Models trained on **Augmented Dataset**, then strictly evaluated against the **Real-World Test Set**.

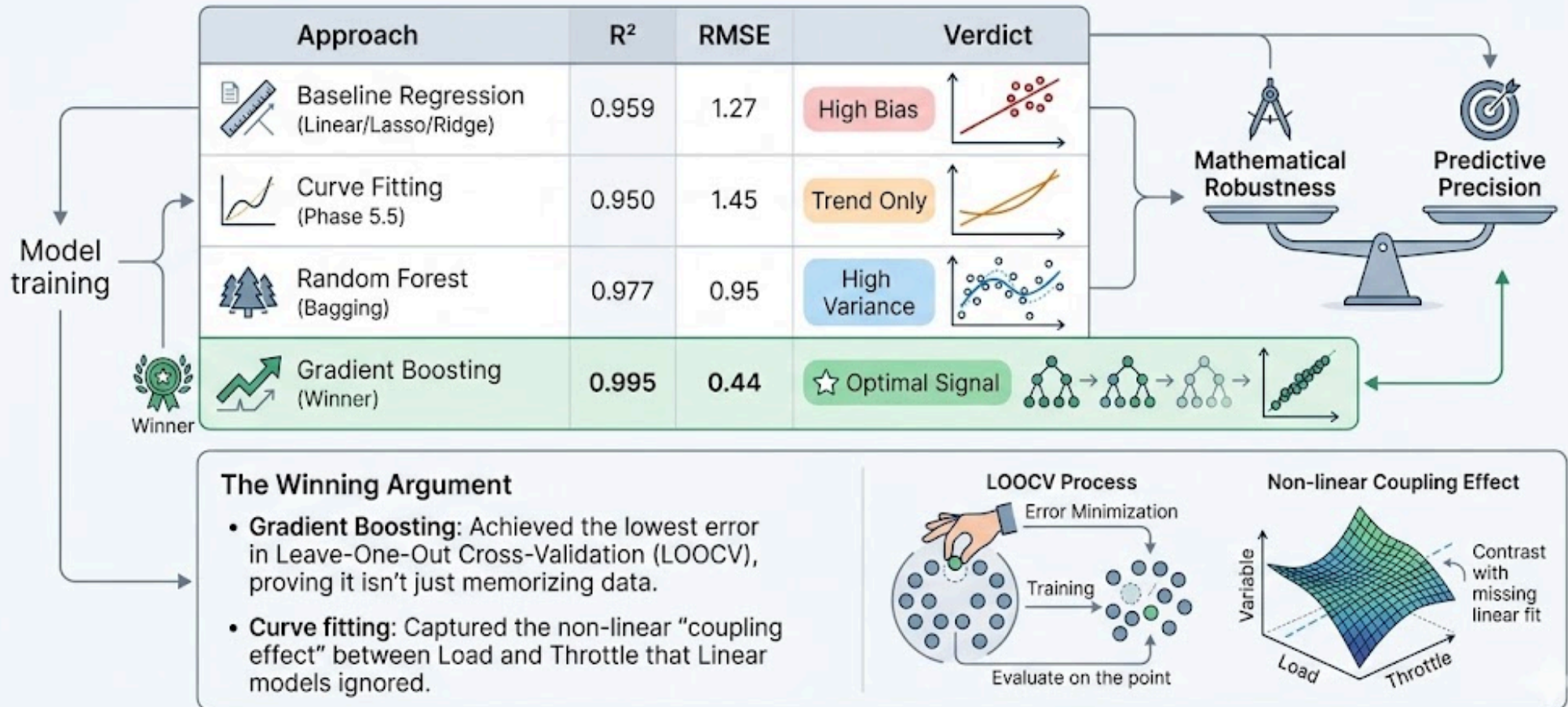
2. Comprehensive Model Validation

Objective: Deploy multiple statistical validation techniques to guarantee model stability, robustness, and prevent overfitting, particularly critical with a small original dataset.



Comparative Analysis & Final Model Selection

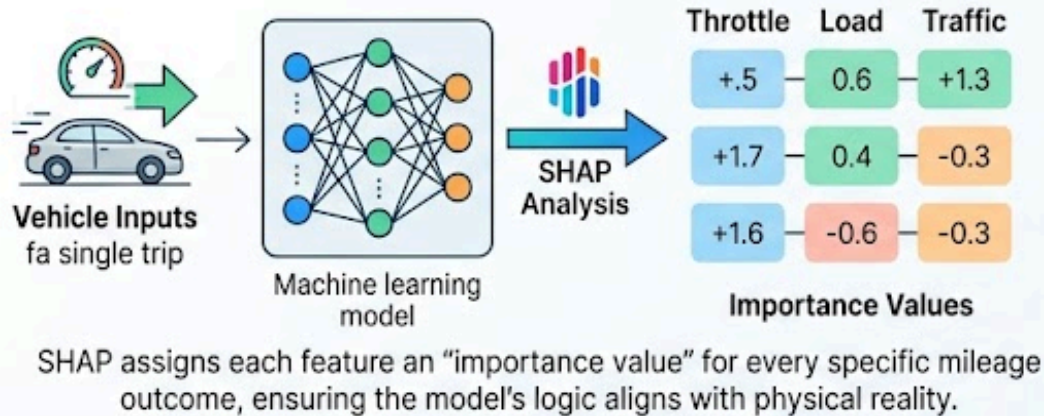
Evaluating Mathematical Robustness vs. Predictive Precision



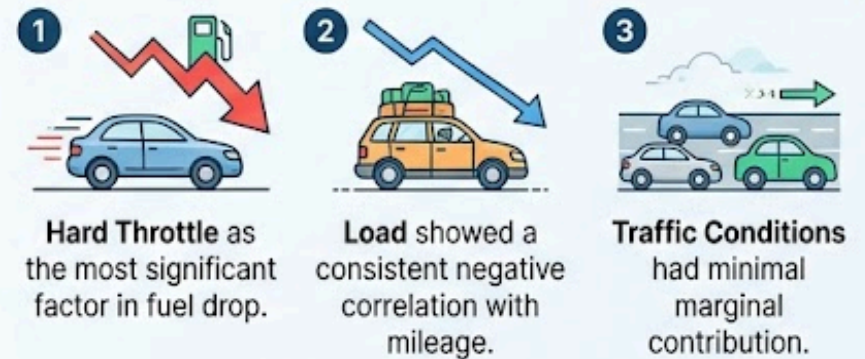
SHAP Analysis

UNDERSTANDING MODEL PREDICTIONS FOR VEHICLE MILEAGE

The Core Idea

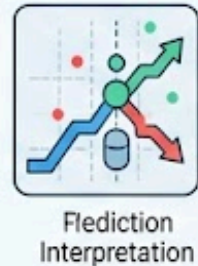


Inferences



Local Interpretation (Force Plots)

Allows us to see fast-regated sampanencing the mieages, and in trofflic cause to nondishing the mileage factormost ereactly in the physical model.

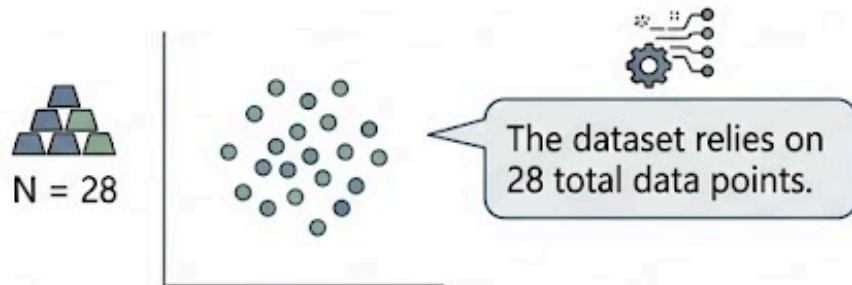


Allows us to see exactly how specific inputs "push" the predicted mileage higher or lower than the base average.

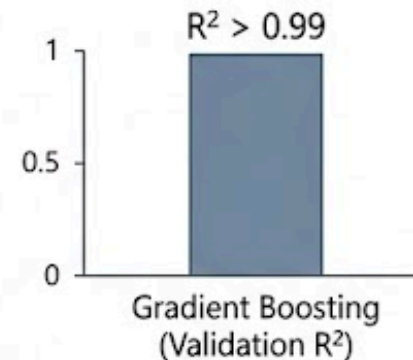
Limitations & Statistical Anomalies

Addressing the risks of small-sample machine learning

Primary Limitation: Data Scarcity

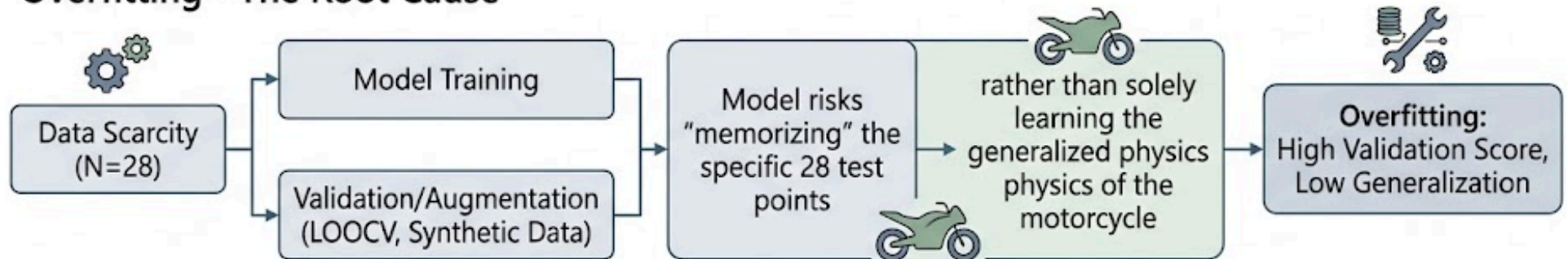


The Statistical Anomaly: "Hyper-Accuracy"



Observation: Models like **Gradient Boosting** reported exceptionally high validation scores ($R^2 > 0.99$).

Overfitting - The Root Cause

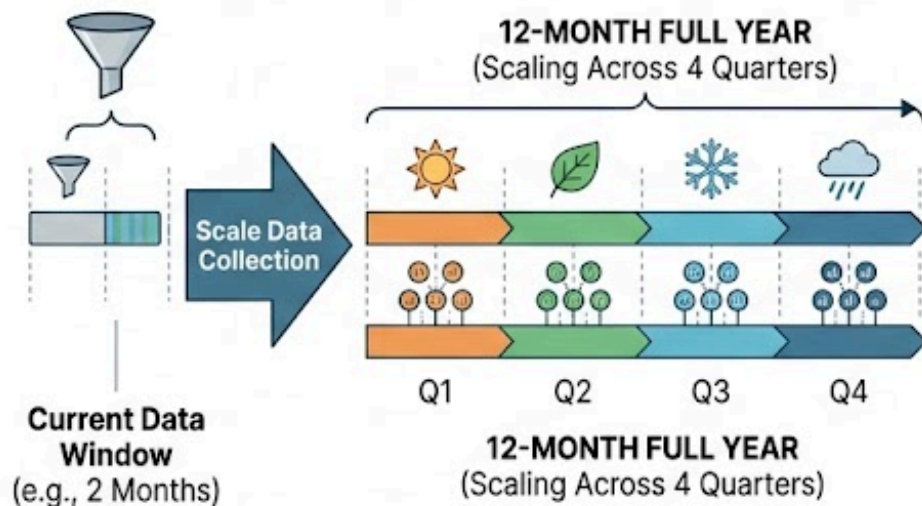


Future Scope & Model Scalability

Expanding data horizons and mass-market application.

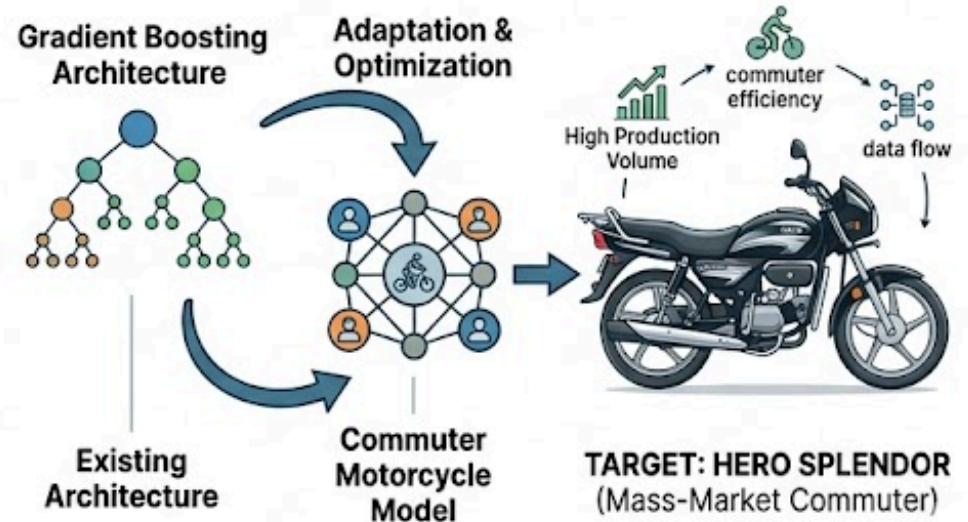
1. Longitudinal Data Strategy (12-Month Horizon)

* Currently, our model is highly accurate but based on a narrow time window. Future efforts require scaling data collection across a full year.



2. High-Volume Commuter Motorcycle Target

We will adapt this Gradient Boosting architecture for high-volume commuter motorcycles, specifically targeting bikes like the Hero Splendor.



Acknowledgments & Credits

With gratitude to those who supported this research.

Academic Guidance & Vision



Prof. Dilip Sharma

Dr. Gunjan Soni

Project Mentorship



Hitesh Sir

Technical Support (Teaching Assistants)



Roshan Boss & Aashutosh Boss

Peer Support & Collaboration



Akshat Kumar

Telemetry, Data & References



STRAVA App (Route & Elevation Tracking)

Bajaj Motors Official Website

Hero MotoCorp Official Website